

Marine biofouling mitigation of PDMS-based network coating with cross-linked contact- and release-active organoalkoxysilane

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ABSTRACT

Poly (dimethylsiloxane) (PDMS) elastomer is one of key polymer resins for fouling release (FR) coatings, but it shows extremely poor antifouling (AF) activity in seawater since bacteria and diatoms could readily attach to surface. Modifying PDMS-based coatings via incorporating antifouling additives is effective for enhancing overall FR and AF activity. In this work, we prepared a well-defined organoalkoxysilane coating additive with quaternary ammonium and benzoisothiazolinone pendent groups, and the synthesized copolymers were covalently conjugated onto PDMS network in a one-pot reaction. AF and FR activities of the modified PDMS coatings were studied against soft fouling microorganisms. When microorganisms come into contact the coating surface, contact-active action mode works, quaternary ammonium groups could cause bacteria cell membrane disruption and microalgae photosynthesis inhibition via positive charge. Unexpectedly, a surprisingly lasting antimicrobial efficiency could be still observed in the neighboring coating, even the top-surface was completely covered with bacteria biofilm. Chemically conjugated benzoisothiazolinone could be sustainably released via side-group hydrolysis, thus, repelling and preventing the planktonic foulings from approaching the coating surface via release-active action mode. The modified PDMS coating with dual antifouling action modes could inhibit the biofilm formation at solid-liquid-air three-phase contact line, and could find valuable applications in marine transportation, water treatment and other antifouling fields.

1. Introduction

Marine biofouling causes detrimental effects on maritime industry and marine economy. Up to now, antifouling coatings have been regarded as one of the most economical and effective approaches to combat the threat from marine biofouling [1–3]. Polydimethylsiloxane (PDMS) elastomer is one of common polymer resins for currently developed fouling release paints, and hard marine foulings (such as adult barnacles and mussels) could be removed when the coated vessel moves or after cleaning procedures [4]. However, PDMS elastomer coatings without any chemical modification still exhibit the unsatisfying resistance to the settlement and releasing of soft fouling organisms, especially for marine diatoms and bacteria, even in dynamic conditions [4,5]. To improve the antifouling activity, zwitterionic, amphiphilic polymers, bactericidal polycations, and/or non-metal biocides have been incorporated into PDMS matrix, to meet the demand of universal antifouling in complex marine environment [6–10].

Ecofriendly bioactive molecules from natural products have been

introduced into PDMS matrix to improve antifouling performance without causing environmental burdens [11–15]. Generally, such discovered antifoulants are only just physically added or chemically bonded into PDMS matrix to perform release- or contact-active antifouling dual action mode. Simultaneously realizing such offensive and defensive action mode is essential for engineering key antifouling materials in complex marine environment [16]. The mutually exclusive paradox, the difficulty of discharging release-active biocides from low-surface-energy PDMS matrix and the trade-off effect between contact-active function and the minimized adhesion force for fouling release ability, are general agreements in antifouling fields [17,18]. In the light of the low-surface-energy character of PDMS resins, the antifoulants embedded into PDMS matrix are difficult to release although the releasing-active biocides are generally not combined with contact-active components since the electrostatic force between coating and fouling organisms contradicted to the fouling release behavior required to the minimized adhesion force [17,19–21]. It is accepted that the combination of hydrophobic PDMS resin and hydrophilic quaternary ammonium

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components could provide promising opportunities to balance the inherent trade-offs and paradoxical surface characters in the same coating system.

In this work, we developed a novel organoalkoxysilane bearing quaternary ammonium and antifoulant pendant groups. Such copolymers with contact- and release-active functions were covalently grafted onto the cross-linked PDMS network as surface-active boundary layer of the antifouling coatings. Bacteria cell membrane disruption and microalgae photosynthesis inhibition were caused by the quaternary ammonium segments, and an unexpected antimicrobial activity was observed in the vicinity of the coating once the coating surface was completely covered with a dead bacteria layer. Excluding the effects from hydrolysis-induced release of antifoulants, the results indicated that the bacteria activated by quaternary ammonium compound might communicate with other neighboring ones, causing retarded bioaccumulation. The modified PDMS elastomer coating could be utilized to inhibit the biofilm formation at solid-liquid-air triple interface, and could find valuable applications in marine transportation, water treatment and other antifouling fields.

2. Results and discussion

2.1. Material preparation and characterization

Ecofriendly 1, 2-benzisothiazolin-3-one (BIT) was selected as the antifoulants owing to its broad-spectrum antibacterial feature [22]. BIT was armed with $-\text{CH}_2\text{OH}$ after reflux with formaldehyde solution at 60°C for 8 h, the purified and dried solid product was termed as BIT-OH.

^1H NMR result in Fig. S1 shows a strong chemical signal assigning to $-\text{CH}_2\text{OH}$. A BIT-based methacrylic acid monomer, *N*-methacryloyloxy methyl benzisothiazolinone (V-BIT), was synthesized by nucleophilic addition between hydroxyl groups and acyl chloride with the aid of trimethylamine (Fig. 1A i). V-BIT and 2-(dimethylamino) ethyl methacrylate (DMAEMA) copolymers (PDMAEMA-*co*-PVBIT) terminated with hydroxyl groups were synthesized, and a free radical statistical copolymerization of the two monomers was conducted using an azo free radical initiator and a thiol chain transfer agent (Fig. 1B). The polymer structure of PDMAEMA-*co*-PVBIT was characterized by ^1H NMR and FTIR spectra (Figs. S2a and S3). PDMAEMA-*co*-PVBIT was protonated by bromobutane under the heat reflux to synthesize PEB-*co*-PVBIT copolymer. As shown in Fig. S3, two peaks at 2824 and 2771 cm^{-1} assigned to $\text{N}-\text{CH}_3$ and $\text{N}-\text{CH}_2$ were disappeared, indicating that the successful synthesis of quaternary ammonium groups. PEB-*co*-PVBIT-IPTES was obtained by a coupling reaction between (3-isocyanatopropyl)-triethoxysilane (IPTES) and PEB-*co*-PVBIT. As shown in Fig. S3, the peaks at around 2870 and 817 cm^{-1} were assigned to stretching and bending vibration of $-\text{CH}_3$ bonds on PEB-*co*-PVBIT, respectively. The peaks at around 1535 and 1086 cm^{-1} were from the stretching vibration of amide groups and $\text{Si}-\text{O}-\text{C}$ bonds, respectively. These results indicated that the coupling reaction between IPTES and PEB-*co*-PVBIT was successfully realized. After hydrolysis and recoupling of siloxane, HO-PDMS-OH could be conveniently coupled with METES and OH-terminated PEB-*co*-PVBIT-IPTES polymers a one-pot reaction. Thus, different contents of PEB-*co*-PVBIT-IPTES (1.0 %, 5.0 %, and 10.0 %) could be connected into the developed PDMS-based crosslinked network coating.

We evaluated the surface wettability change of the coating (with a

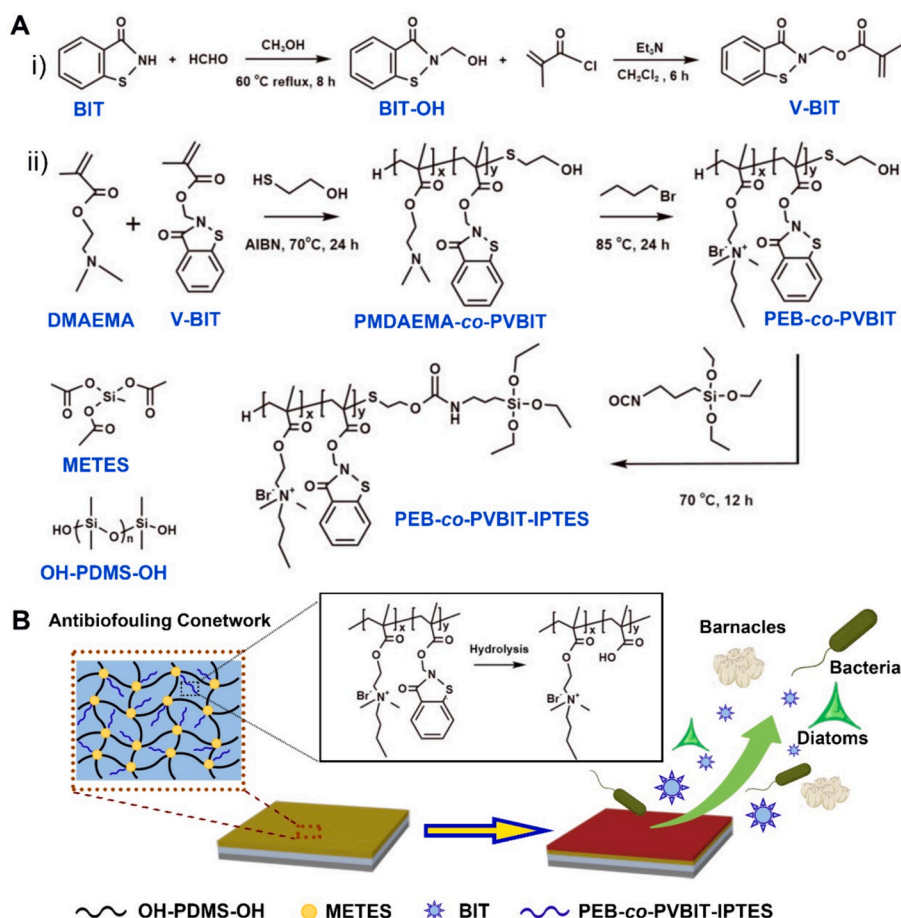


Fig. 1. A) Schematic diagram of synthesis of V-BIT antifoulant precursor (i) and organoalkoxysilane antifouling additive (ii). B) Schematic illustration of the PDMS-based coating with surface-active organoalkoxysilane.

dry thickness of around 300 μm) via monitoring the water contact angles when varied PDMS-based coatings were soaked into seawater for 105 days. As shown in Fig. 2A, water contact angles of varied material surface were over 105° . After seawater submersion, the values of water contact angle gradually reduced with increasing submersion time and finally were remained unchanged. Especially, the water contact angle of PDMS- $\text{P}_{10.0}$ coating surface was reduced from $107.3 \pm 4.2^\circ$ to $82.0 \pm 2.7^\circ$. Since the BIT pendant groups were hydrolyzed and evolved into hydrophilic acrylic moieties, water contact angles of the coating were evidently reduced. Whereas, water contact angle of unmodified PDMS surface was maintained unchanged. In addition, it was obvious that the water contact angles of varied coatings were reduced as increasing the contents of PEB-co-PVBIT (Fig. 2A). The surface free energies of PDMS-based coatings were determined on the basis of typical Owen-Wendt approach through measuring water and diiodomethane contact angles.

As shown in Fig. 2B, the surface free energies of all the coating were still below $30 \text{ mJ}/\text{m}^2$, indicating that the modified coating still possessed remarkable fouling-release ability. To confirm that the increased coating hydrophilicity is owing to the hydrolysis of BIT pendant group, the changed of chemical structure of PEB-co-PVBIT before and after hydrolysis were studied. As shown in FTIR spectra in Fig. S4, the absorption peak at around 848 cm^{-1} assigning to COO^- group was present, and the absorption peak at about 2854 cm^{-1} assigning to C—H bond and the peak at around 1562 cm^{-1} belonging to the benzene rings were disappeared on PEB-co-PVBIT after water hydrolysis. In addition, the peak at 5.7 ppm chemical shift was disappeared, further suggesting the hydrolysis of antifoulant side groups (Fig. S5). These results indicated that BIT was released by ester hydrolysis, and the hydrophilic carboxyl groups were remained on the polymer backbone. The varied surface wettability and surface free energy should be due to the exposed

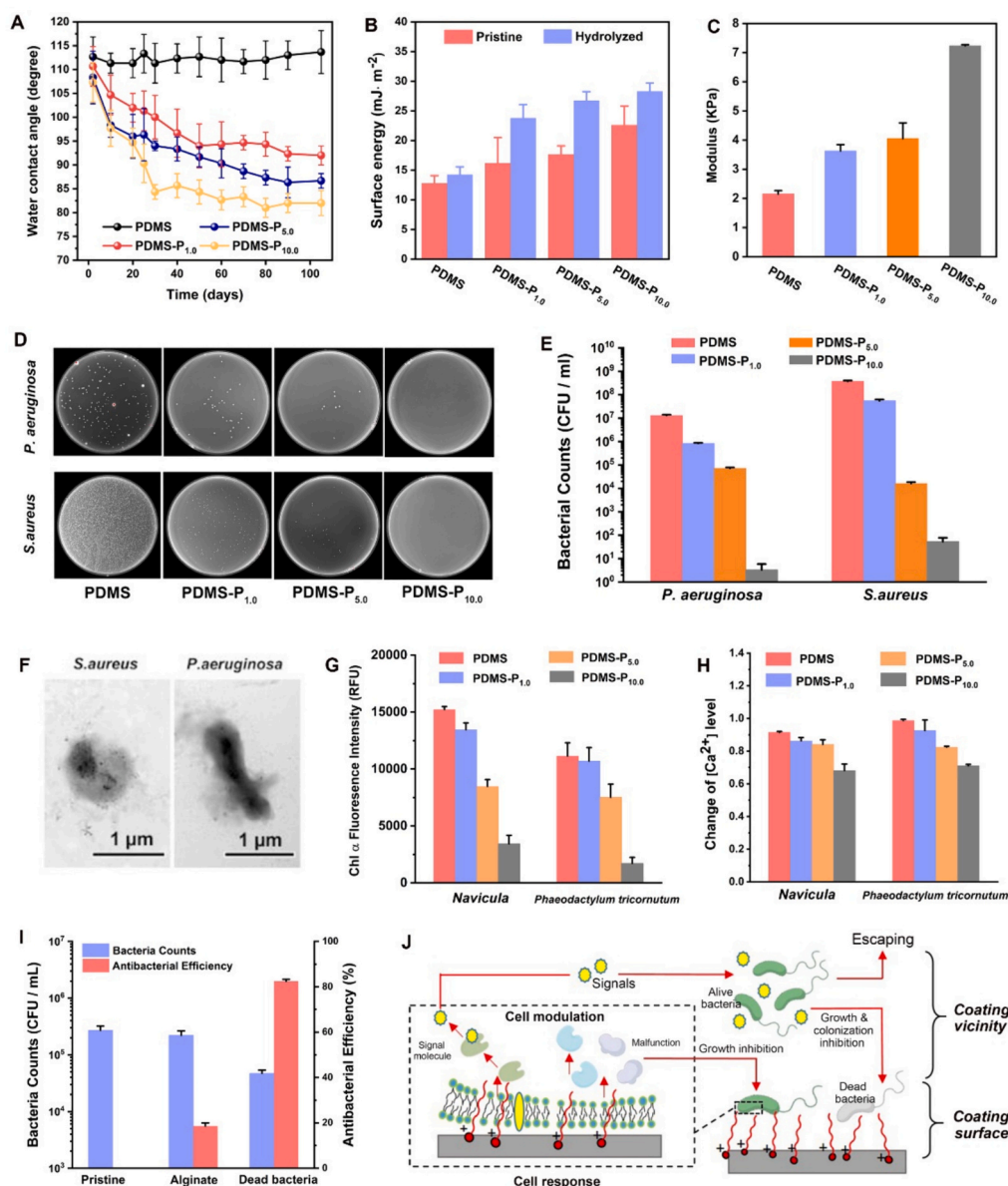


Fig. 2. A) Water contact angle variation of PDMS-based coatings with increasing seawater-immersion time. B) Surface energy of PDMS-based coatings before and after hydrolysis. C) Bulk modulus of varied PDMS-based coatings. D) Photographs of *P. aeruginosa* and *S. aureus* colony after incubation with varied PDMS-based coating. E) Bacterial CFU after 4 hour incubation with varied PDMS-based coating. F) TEM images of two bacteria after incubation with PDMS- $\text{P}_{10.0}$. G) The relative fluorescence units (RFU) of the extracted chlorophyll and H) Cellular Ca^{2+} level in two marine diatom cells adhered on PDMS-based coatings. I) Bacteria counts and antibacterial efficiency of PDMS- $\text{P}_{10.0}$ completely covered with abiotic alginate and *S. aureus* fouling layers. J) Schematic illustration of the antimicrobial behaviors at modified PDMS coating surface and in the surround coating.

hydrophilic carboxyl groups caused by BIT releasing after water hydrolysis.

Tensile stress-strain curves and bulk modulus of varied PDMS-based coatings were further investigated. As shown in Fig. S6, introducing of PEB-co-PVBIT-IPTES organoalkoxysilane could increase the mechanical strength of polymer coating and reduce its toughness. The tensile strength and bulk modulus of the coating were increased (Fig. 2C), and the elongation at break was decreased with increasing the contents of organoalkoxysilane, which might be due to the harder polymer chain of PEB-co-PVBIT-IPTES. In turn, it can be reflected that the synthesized organoalkoxysilane possess a reasonable compatibility with PDMS.

2.2. Contact-active antimicrobial performance of PDMS-based coating

Considering the quaternary ammonium groups in PEB-co-PVBIT-IPTES, the rate of survival for two model bacteria *P. aeruginosa* and *S. aureus*, incubated with varied PDMS-based coatings was used to assess the bactericidal performance of different surfaces via contact-killing action mode. As expected, the modified PDMS shows antibacterial activity toward the two bacteria strains, and the bacteria colony numbers were reduced with increasing the contents of PEB-co-PVBIT-IPTES in the coating (Fig. 2D). As shown in Fig. 2E, the modified PDMS coating exhibited antibacterial activity against both two bacteria strains in a dose-dependent manner. PDMS-P_{10.0} displayed evident bactericidal effect, and over 99.9 % bacteria could be killed after the bacteria were in contact with PDMS-P_{10.0} coating surface after 4 h co-incubation. TEM images of two bacteria attached on PDMS-P_{10.0} coating surface in Fig. 2F showed that the cell membranes were ruptured after being in contact with the modified PDMS coating. The bactericidal performance was ascribed to the contact-killing action mode from quaternary ammonium moieties in organoalkoxysilane.

The modified PDMS coating with surface-active organoalkoxysilane exhibited antibiofilm activity for marine diatoms *Navicula trivialis* and *Phaeodactylum tricornutum*. The amounts of biomass were represented by relative fluorescence units (RFU) obtained from extracting chlorophyll from the cells. As shown in Fig. 2G, the fluorescence intensity values associated with two marine diatom biomass were significantly decreased with increasing contents of bioactive organoalkoxysilane, indicating that the modified PDMS coating possessed an important inhibitor effect on the photosynthetic activity. To investigate the underlying algacide mechanisms, intracellular Ca²⁺ concentration levels were detected for the attached marine diatom cell on varied coatings. As shown in Fig. 2H, intracellular Ca²⁺ concentration levels of both two marine diatoms attached on the modified PDMS coatings were all decreased compared with the pristine PDMS surface. The reduced Ca²⁺ concentration level might cause the instability of the cell membrane, making cell membrane readily to be attacked because Ca²⁺ plays a crucial role in maintaining the integrity of phospholipid membrane structure by crosslinking with the negatively charged head groups of lipids [23–25].

As mentioned, the modified coatings with bioactive organoalkoxysilane exert their antibacterial activity and inhibitory effect on photosynthetic ability for marine diatoms based on the contact-killing action mode caused by the quaternary ammonium segments. For the classical contact-killing strategy, the bioactivity of the coating is severely restricted to the surface, that is, the coating tends to lose its bioactivity when the coating surface was completely covered. We investigated the bactericidal performance around the prefouled PDMS-P_{5.0} surface after deposition of two model fouling ingredients on coating surfaces: abiotic alginate and *S. aureus*. We investigated the bacteria counts of *S. aureus* and corresponding antibacterial efficiency of the completely covered coating surfaces. As shown in Fig. 2I, the antibacterial activity of PDMS-P_{5.0} coating surface was significantly reduced when the coating surface was thoroughly covered with biotic alginate, and the antibacterial efficiency was only 18.6 %. In contrast, the coating surface covered with around 25 μm thickness *S. aureus* biofilm still

exhibited a persistent bactericidal performance, and the antibacterial efficiency maintained at 82.4 % (Fig. S7). It is noted that PDMS-P_{5.0} coating surface could be thoroughly covered because $\sim 25 \mu\text{m}$ thickness model cake layer could offer a sufficient coverage of the alkyl chains of quaternary ammonium components. The proposed mechanism of PDMS-P_{5.0} coating covered with *S. aureus* biofilm was shown in Fig. 2J. Quaternary ammonium component through disrupting cell membrane integrity could cause cell response including cell respiratory enzyme inhibition and proton motive force dissipation, thus, might triggering SOS feedback through cell regulation or signaling [26]. The cell communication between the attacked bacteria remained on coating surface and those in surround coating might result in the inhibition of cell growth and colonization [27]. It was previously reported that the microorganisms could adopt contacting-associated cell signaling to restrict cell growth and colonization. In addition, the bactericidal activity nearby the coating surface might be associated with the established programmed cell death pathway with quorum sensing peptides in a toxin-antitoxin system such as *mazEF* [28,29].

2.3. Sustained-release of BIT antifoulant from PDMS-based coatings

The quaternary ammonium moiety could inactivate the bacteria and disrupt photosynthesis procedure of diatoms attached on coating surface, indicating that the coating possessed favorable contact active antifouling action mode. Owing to the molecular character of methacrylate, BIT segment in organoalkoxysilane could be released by hydrolysis of side groups in natural seawater. BIT molecule is water-soluble, low molecular weight, and ecofriendly biocide in packaging and painting industries for its wide-spectrum antimicrobial ability and fast degradation rate in aqueous environment [30–32]. The bioactivity of BIT molecule arises from the binding of N-S bond in heterocyclic ring to cellular protein [33]. However, the release rate of BIT in coatings is challenging to control, because such small molecule with low molecular weight tends to migrate to the coating top surface in the coating formation and curing procedure. To control the release rate of BIT in coatings and prolong its service life, BIT groups is chemically grafted to main chain of PEB-co-PVBIT polymer, and the self-polishing PDMS-based antifouling paints were also prepared with PEB-co-PVBIT copolymer as antifouling additives. As shown in Fig. 3A, UV-vis spectra of standard BIT and hydrolysis product collected in seawater were almost identical, indicating that the released product from polymer hydrolysis may well be BIT-based biocide. The release behavior of BIT hydrolysis product was shown in Fig. 3B and C. All the modified PDMS coatings could sustainably release BIT for at least 105 days. The BIT releasing rate was evidently accelerated with increasing the content of PEB-co-PVBIT in the modified PDMS coating. The releasing of BIT further improves the antifouling activity of the coating against marine diatoms and bacteria in natural seawater, and the released BIT from the modified coating surface could repel and inhibit foulings from approaching the coating surface via release-killing action mode.

2.4. Antifouling and fouling release performance of PDMS-based coatings

Considering the contact- and release-antifouling action mode, we evaluated the antifouling and fouling release performance of series of PDMS-based coatings against bacteria and marine diatoms in static conditions. The formation of bacteria biofilm is the crucial step for developing marine biofouling, thus, the formed biofilm could provide favorable conditions for settlements and adhesion of other marine fouling organisms [34,35]. Inhibition of bacteria adhesion, no matter alive and dead bacteria, is one of crucial criteria for characterizing antifouling ability of a coating. As shown in Fig. 3D, many bacteria was still observed on PDMS and the modified PDMS coatings before water jet, although the quaternary ammonium groups in the modified PDMS coating surface could kill the attached bacteria when bacteria was in contact with the coating. The attached bacteria was completely removed

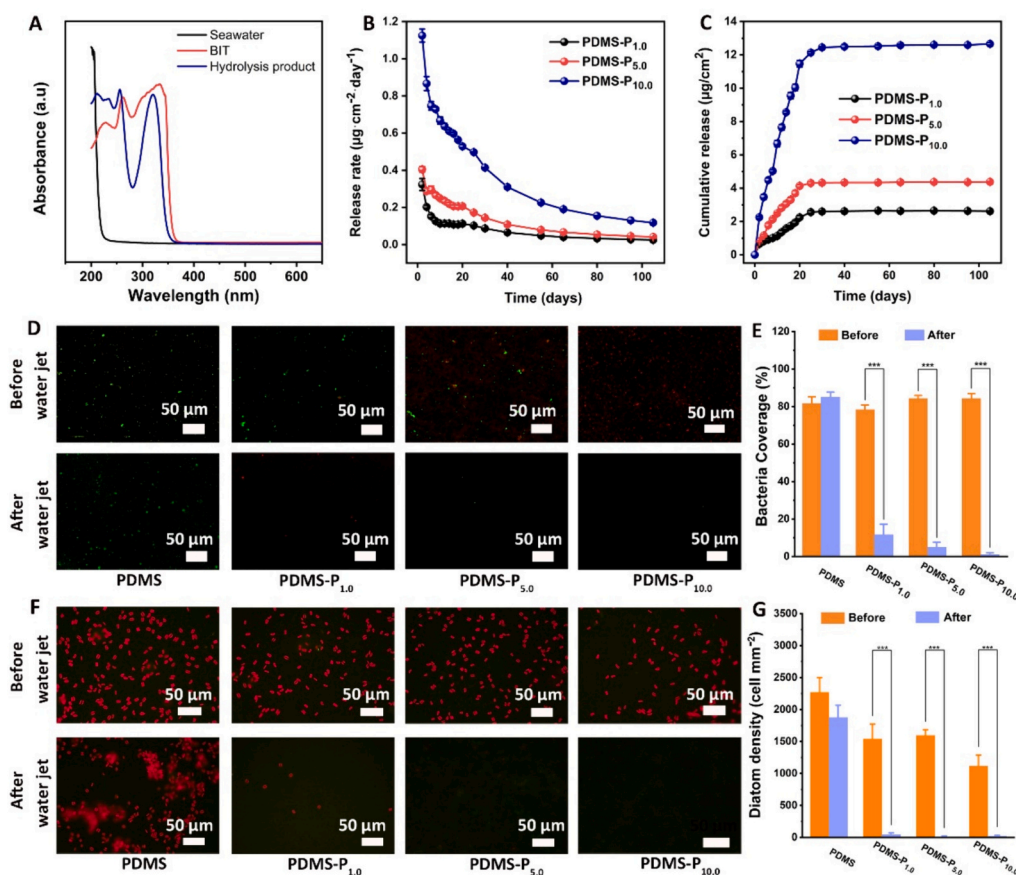


Fig. 3. A) UV-vis spectra of standard BIT and hydrolysis products. B) Release rate and cumulative release (C) of BIT from PDMS-based coatings in seawater. D) Representative fluorescence microscopy images of attached *P. aeruginosa* on PDMS-based coatings before and after water jet. The bacteria were dyed with LIVE/DEAD assay kit. E) Bacteria coverage on PDMS-based coatings before and after water jet. F) Representative fluorescence microscopy images of *Navicula trivialis* on PDMS-based coatings before and water jet. G) *Navicula trivialis* density on the coatings before and after water jet.

from the modified PDMS coating surface by the water-jetting process, but a considerable amount of bacteria was still could be observed on pristine PDMS surface even after water jet. After water jet, the bacterial coverage of 85.11, 94.07 and 98.81 % were reduced for PDMS-P_{1.0}, PDMS-P_{5.0} and PDMS-P_{10.0} coating surface (Fig. 3E), respectively. This response demonstrated that the modified PDMS coatings not only had good antifouling activity via inducing bacteria death, but had a good fouling release character toward *P. aeruginosa* adhesion. The fouling release ability toward marine bacteria was significantly enhanced with increasing the contents of surface-active organoalkoxysilane in the modified PDMS coatings.

It is accepted that PDMS-based coating possessed the poor resistance to the settlement and adhesion from soft marine fouling microorganisms [36]. A soft slime layer could still be formed on PDMS-coated ship hull at high navigation speeds of 30 knots [37,38]. *Navicula trivialis* as one of representative marine fouling diatoms was selected to evaluate the antifouling ability against settlement of marine microorganisms. As shown in Fig. 3F, a considerable amount of *Navicula trivialis* diatom cell was attached on the pristine PDMS surface even after the water-jetting process. In contrast, *Navicula trivialis* adhesion on the modified PDMS coating was significantly reduced after the water-jetting process. After water jet, the diatom cell coverage of 97.12, 99.36 and 97.98 % were reduced for PDMS-P_{1.0}, PDMS-P_{5.0} and PDMS-P_{10.0} coating surface (Fig. 3G), respectively. The strong hydrophobic interactions resulted in the strong adhesion between PDMS surface and marine diatoms, thus, *Navicula trivialis* preferentially attached on the pristine PDMS coating surface even after water jet. While for the modified PDMS coating surface, the released BIT antifoulant caused by hydrolysis of side groups

could repel and prohibit *Navicula trivialis* from coming close to the coating surface, and the hydrophilic residue could reduce the diatom attachment on PDMS-based coating surface.

The pseudobarnacle adhesion test was utilized to study the fouling release ability of PDMS-based coatings with a dry thickness of around 300 μm , and the bond pull-off strength of the coatings was measured for evaluating the interfacial adhesion strength between the primer and the coating. The schematic illustration for measuring the shear and the tensile strength was shown in Fig. 4A. Fig. 4B exhibited the pseudobarnacle adhesion comparison on varied coatings, and it can be observed that the shear force strength on the modified PDMS coating was obviously lower than that on the pristine PDMS coating. It is believed that a lower removal strength represented a better fouling release ability, thus, the attached fouling organisms could be easily removed on the modified PDMS coating surface. After hydrolysis, the shear strength of PDMS-P_{10.0} coating was reduced from 0.17 to 0.06 MPa, indicating that the hard fouling organisms could be easily removed by the hydrodynamic forces. The adhesion strength of a coating with the substrate is crucial criteria for evaluating the mechanical strength of antifouling coating, because sufficient interfacial adhesion between the substrate and coating is highly demanded for use in rigorous marine environment. The interfacial adhesion was evaluated through measuring the tensile strength of the coatings with epoxy panel substrate. As shown in Fig. 4C, silicone modification lead to the decrease in interfacial adhesion strength of the coatings, and a descending trend was observed when increasing the synthesized nonpolar organoalkoxysilane in the PDMS-based coatings. The interfacial adhesion strength was decreased after hydrolysis of side groups. This might be attributed to the

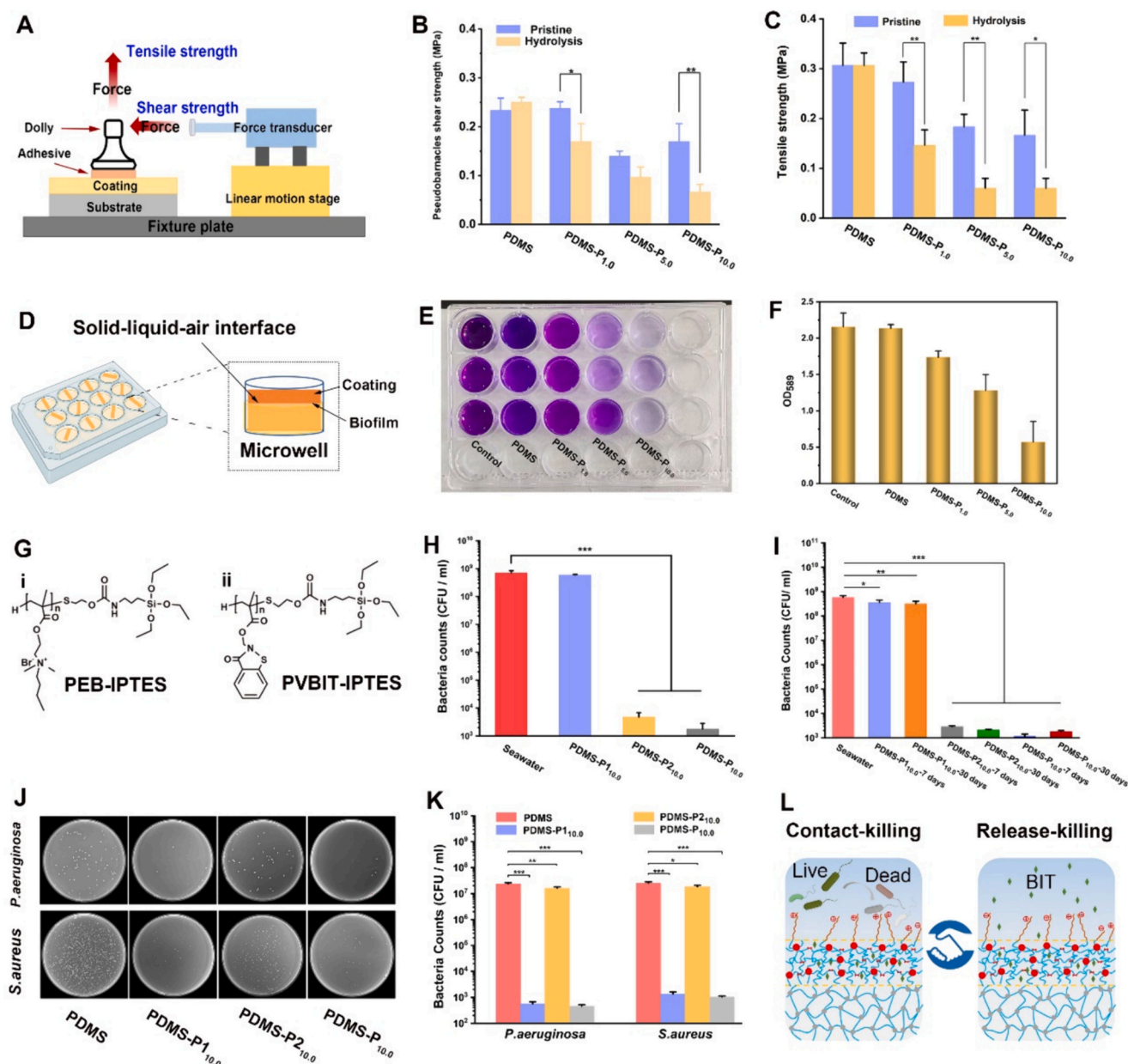


Fig. 4. A) Schematic illustration of the tests for *Pseudobarnacle* shear strength and tensile strength measurements. B) *Pseudobarnacle* shear strength of PDMS-based coating before and after hydrolysis. C) Tensile strength of PDMS-based coating before and after hydrolysis. D) Schematic illustration for the test to investigate bacteria adhesion at solid-liquid-air triple-phase interface. E) PAO 1 biofilm grown on the PDMS-based coatings stained with crystal violet, and the detected OD₅₈₉ values from optical density test (F). G) The chemical formula of PEB-IPTES and PVBIT-IPTES. H) The counts for the attached *P. aeruginosa* on modified PDMS coating with varied organoalkoxysilane. The addition weight of organoalkoxysilane was all adjusted to 10 wt%. I) Count variations of the attached *P. aeruginosa* on modified PDMS coating with varied organoalkoxysilane in 7 and 30 days. J) Photographs of *S. aureus* and *P. aeruginosa* colonies after incubation with the modified PDMS coating with different additives. K) Bactericidal activity of the modified PDMS coating with different additives. L) Schematic illustration of contact-killing and release-killing mode of the modified PDMS coating.

coating swollen caused by the water absorption by hydrophilic residues on the coating surface.

2.5. Solid-liquid-gas interfacial antifouling performance

Aerobic marine microorganisms, such as bacteria and marine diatoms, and eukaryotic fouling species, such as microalgae and barnacle cyprids, are prone to accumulate at solid-liquid-air triple interface [39–41]. The preferential settlement and accumulation of marine fouling species at solid-liquid-air triple interface is mediated by the convenient access to nutrients and oxygen. Consequently, the biofouling at solid-liquid-air interface causes great troublesome. However,

currently developed antifouling systems mainly focus on prevent biofouling on completely submerged material surface [42], while, the severe biofouling at three-phase contact line should require more research attention. Considering the structure feature of the synthesized organoalkoxysilane, we investigated the biofouling inhibition performance of our modified PDMS coating at three-phase contact line. Schematic diagram to culture biofilm at solid-liquid-air triple interface using the modified PDMS coating was shown in Fig. 4D. We selected *Pseudomonas aeruginosa* strain PAO1 with abundant Type IV pili appendages to study the inhibition ability of PDMS-based coating for preventing biofilm formation at three-phase contact line. Fig. 4E and F show the results of PAO 1 biofilm formation on the varied modified

PDMS coatings, it can be observed that the absorbance at 589 nm were all reduced for the varied modified PDMS coatings. The remarkable antifouling ability of the modified PDMS coating at three-phase contact line was attributed to the contact-killing mode at solid-air interface and release-killing mode at solid-liquid interface.

2.6. Antifouling contribution of contact- and releasing-active modes for the as-prepared organoalkoxysilane

We synthesized PEB-IPTES and PVBIT-IPTES organoalkoxysilane to study the contribution of contact- and release-killing mode in the modified PDMS coating (Fig. 4G). PEB-IPTES and PVBIT-IPTES was covalently grafted into PDMS network, respectively. The percentage of these two organoalkoxysilane in the coatings was both adjusted to 10 %. The modified PDMS coating with PEB-IPTES was termed as PDMS-P1_{10.0}, while the coating with PVBIT-IPTES was represented as PDMS-P2_{10.0}. We investigated the antifouling performance against bacteria adhesion on completely submerged coating surface, as shown Fig. 4H, the attached bacteria counts on PDMS-P2_{10.0} and PDMS-P1_{10.0} coating surface were both significantly reduced. Moreover, such two coating exhibited long-last antiadhesion ability at least 30 days (Fig. 4I). The released BIT antifoulant groups caused by hydrolysis of side groups could repel and prevent bacteria from approaching the coating surface, thus, the attached bacteria counts were decreased. The presence of quaternary ammonium fraction can improve the wide-spectrum antibacterial ability of the coating, thus, PDMS-P1_{10.0} and PDMS-P2_{10.0} coating surface exhibited remarkable bactericidal performance against *P. aeruginosa* and *S. aureus* (Fig. 4K). Thus, the coexistence of contact- and release-killing mode in the modified PDMS coating could either inactivate foulings on the coating surface, or the released BIT could repel and prevent foulings from approaching the coating surface (Fig. 4L).

3. Conclusion

In summary, this work provides valuable insights for further investigation in designing AF and FR coating formulations based on dual functional organoalkoxysilane bioactive additives with quaternary ammonium and BIT pendant groups. AF and FR activities against soft microorganisms were enhanced with increasing the contents of the synthesized copolymers. The modified PDMS coating could lead to membrane disruption and microalgae photosynthesis inhibition for the fouling organisms attached on the coating surface, while continually releasing BIT for over 105 days to prohibit planktonic bacteria and marine diatoms from coming close to the coating surface. An unexpected 82.4 % antimicrobial efficiency was still be observed in the vicinity of the modified PDMS coating, even the top-surface was completely covered with 25 μm thickness *S. aureus* biofilm. The modified PDMS coating with contact- and release-active action mode could significantly restrain the biofilm formation at solid-liquid-air three-phase contact line, highlighting the effectiveness of our synthesized surface-active organoalkoxysilane coating additive. The modified PDMS coating exhibited a simultaneously enhanced AF and FR abilities, and a long-lasting antifouling effect. This work provides a great inspiration to develop fouling-resistant PDMS coating via covalent incorporation of novel antifouling coating additive in resin matrix.

CCRediT authorship contribution statement

Yihan Sun: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Data curation. **Haibin Zhao:** Investigation. **Yuanyuan Shen:** Methodology. **Peng Wang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2024.108743>.

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